The Quantum Compass Model

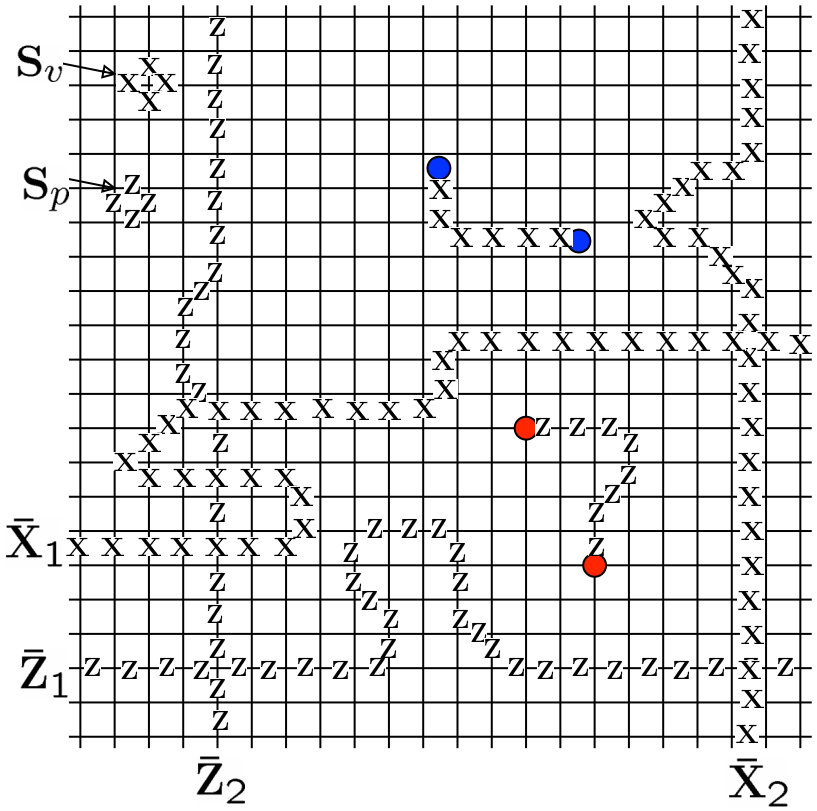


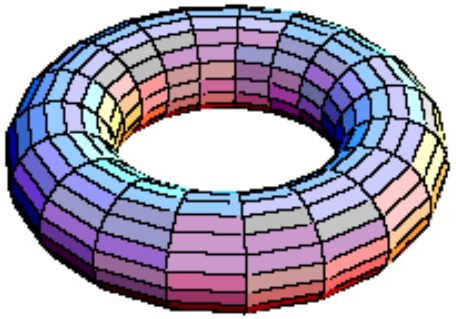
Dave
Bacon

Department of Computer Science & Engineering
University of Washington

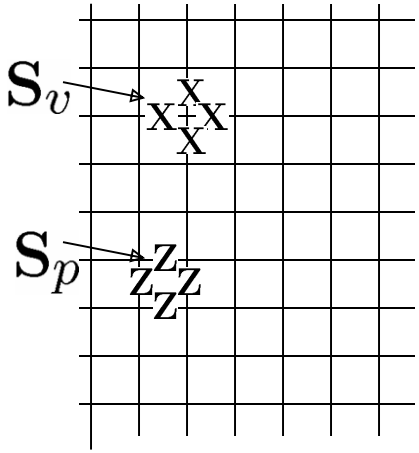
Kitaev's Idea

Are there many-body quantum systems which can be used to robustly store quantum information and keep this information coherent during fault-tolerant computation with this information?





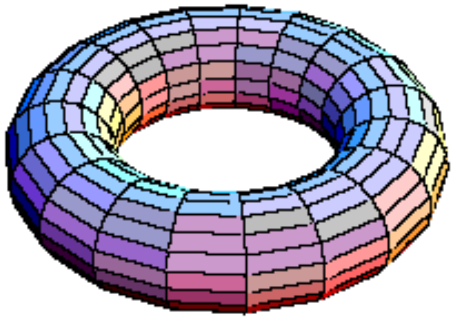
Kitaev's Toric Codes



Qubits on the edges of a lattice on a torus

(torus condition can be

Plaque operators:



$$S_p = \prod_{e \in p} Z_e$$

Vertex operators:

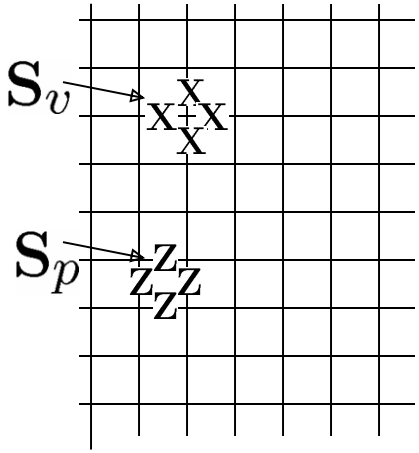
$$S_v = \prod_{e \in v} X_e$$

Stabilizer code: plaque and vertex opertors commute

degenerate +1 eigenspace is quantum error correcting code

$$H = -J(\sum_p S_p + \sum_v S_v)$$

Kitaev's Toric Codes



$$H = -J(\sum_p S_p + \sum_v S_v)$$

Energy spectrum has gap between ground and first excited states

degeneracy is lifted by perturbations only at nth order

Can be used to robustly store quantum information in ground state degeneracy if the temperature is less than the gap.

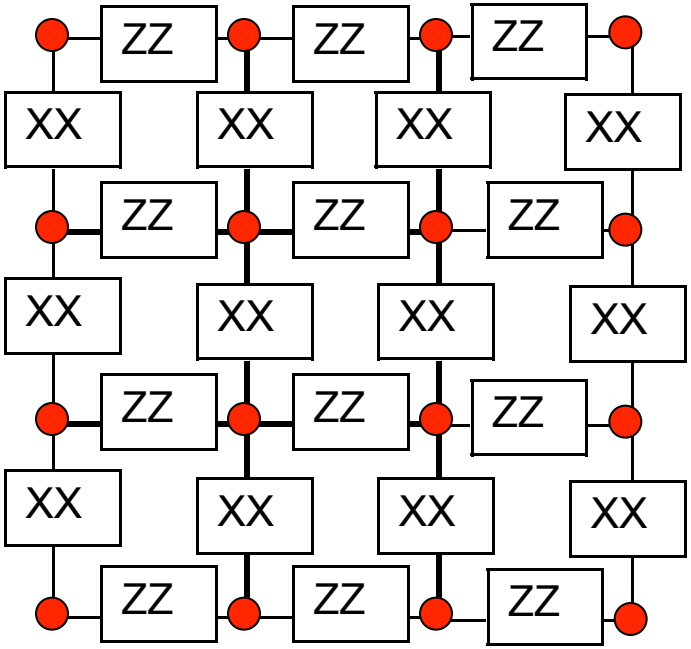
Something I worry about: if you enact operations on such a system, you're errors will become thermal excitations and will disorder the quantum information

Quantum Compass Model

[B. Douçot, M. V. Feigel'man, L. B. Ioffe, and A. S. Ioselevich 05]

[Dourier, Becca, Mila 05]

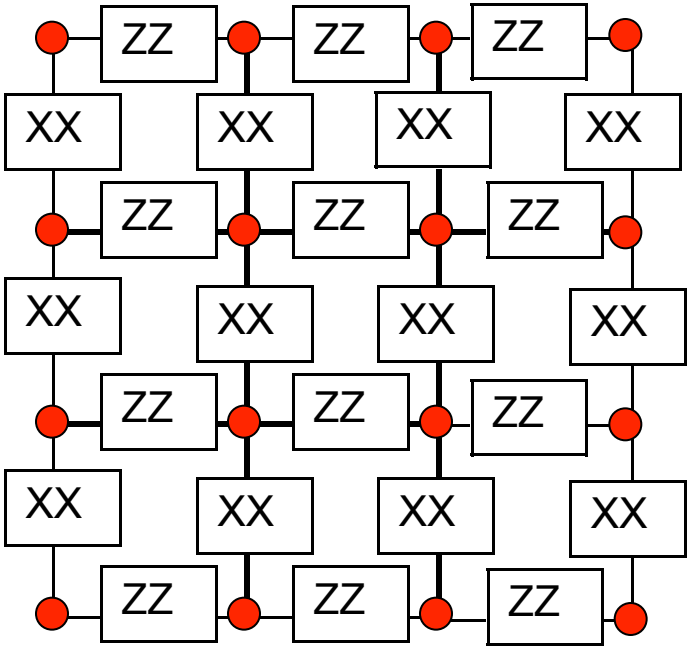
n by *n* square lattice



$$H = -\sum_{i,j=1}^{n} J_X X_{i,j} X_{i+1,j} + J_Z Z_{i,j} Z_{i,j+1}$$

Quantum Compass Model

n by *n* square lattice



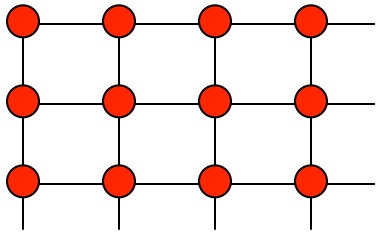
1. The quantum error correcting subsystem

2. Gapped or not gapped?
   Green's function quantum monte carlo

3. More interesting systems…fodder for you all?

Flatland

n by n cubic lattice of qubits:



$$X_{2,2} = \left[\begin{array}{cccc|cccc} 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array}\right]$$

X Z

$$Z_{2,1} = \left[\begin{array}{cccc|cccc} 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array}\right]$$

$$Y_{2,1} = i \left[\begin{array}{cccc|cccc} 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array}\right]$$

generic Pauli group operator on cubic lattice:

$$P = i^k \left[\begin{array}{c|c} A & B \end{array}\right]$$ A, B n by n 0,1 matrices

X Columns, Z Rows

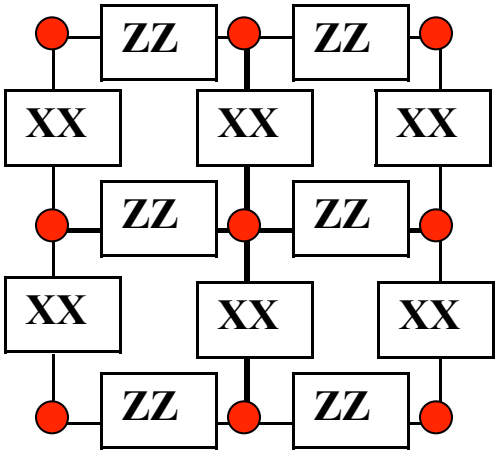
X Z

$$\left[\begin{array}{c|c} A & B \end{array}\right]$$

$\mathcal{T} = \{c\left[\begin{array}{c|c} A & B \end{array}\right]$ | A has an even # of 1′s in columns

B has an even # of 1′s in rows$\}$

$$c\left[\begin{array}{cccc|cccc} 0&1&0&0&1&1&1&1 \\ 0&1&0&0&0&0&0&0 \\ 0&0&0&1&0&0&0&0 \\ 0&0&0&1&1&1&0&0 \end{array}\right] \in \mathcal{T}$$

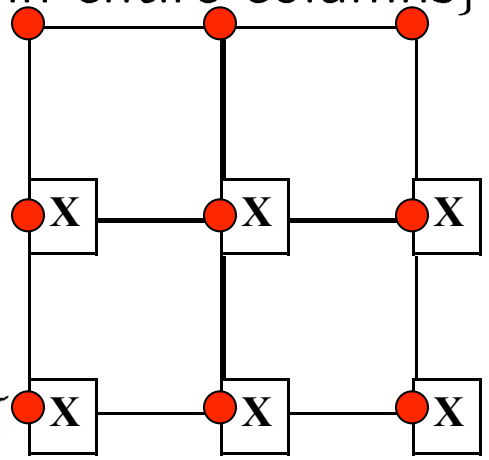


$\mathcal{S} = \{\left[\begin{array}{c|c} A & B \end{array}\right]$ | A has an even # rows with 1′s in entire row

B has an even # columns with 1′s in entire columns$\}$

$\mathcal{S} \subset \mathcal{T}$

$$\left[\begin{array}{cccc|cccc} 1&1&1&1&1&1&0&0 \\ 1&1&1&1&1&1&0&0 \\ 0&0&0&0&1&1&0&0 \\ 0&0&0&0&1&1&0&0 \end{array}\right] \in \mathcal{S}$$



All elements of $\mathcal{S}$ commute with all elements of $\mathcal{T}$

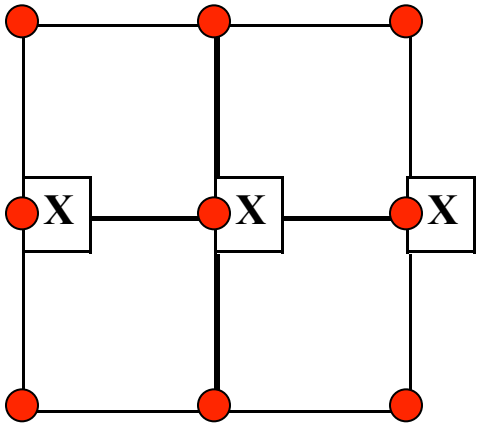
X Columns, Z Rows

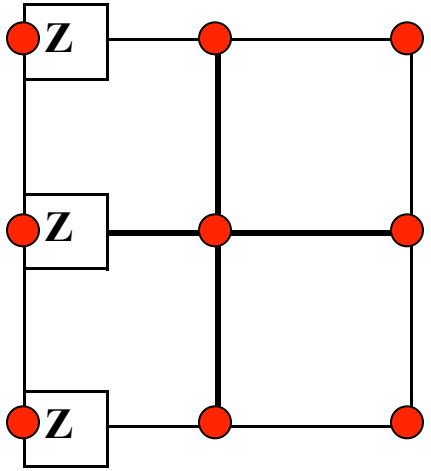
$$\mathcal{L} = \{c\left[\begin{array}{c|c} A & B \end{array}\right] \mid A \text{ has an odd \# rows with } 1\text{'s in entire columns}$$

B has an odd # colums with 1's in entire rows$\}$

$$c\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \in \mathcal{L} \qquad \mathcal{L} \notin \mathcal{T}$$

Elements of $\mathcal{L}$ commute with all elements of $\mathcal{T}$





A Little Representation Theory

$\mathcal{S} = \{ \left[\, A \, \middle| \, B \, \right] \;\; | \;\; A$ has an even # rows with 1's in entire row

B has an even # columns with 1's in entire column$\}$

$\mathcal{S}$ commutes with all of $\mathcal{T}$ and $\mathcal{L}$

Label subspaces by $S = \{ \pm 1$ eigenvalues of elements of $\mathcal{S} \}$

$$\mathcal{H} = \bigoplus_{s \in S} \mathcal{H}_T \otimes \mathcal{H}_L$$

$\dim = 2^{(n-1)^2}$ $\qquad$ $\dim = 2$

Elements $T \in \mathcal{T}$ act as

$$T = \bigoplus_{s \in S} T_s \otimes I$$

Elements $L \in \mathcal{L}$ act as

$$L = \bigoplus_{s \in S} I \otimes L_s$$

Elements $S \in \mathcal{S}$ act as

$$S = \bigoplus_{s \in S} \pm I \otimes I$$

We will encode a single qubit of in $\mathcal{H}_L$

Quantum Error Correcting

$$\mathbf{S}\mathcal{H} = \bigoplus_{s \in S} \mathcal{H}_T \otimes \mathcal{H}_L \mathbf{s}$$

Initialize state over subspace labeled by all $s = +1$ eigenvalues of $S \in \mathcal{S}$ as

$$\rho_{arb} \otimes |\psi\rangle\langle\psi|, \; |\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

Error correction for error E need only protect subsystem

$\mathcal{S} = \{\left[\begin{array}{c|c} A & B \end{array}\right] \;|\; A$ has an even # rows with $1'$s in entire row

B has an even # columns with $1'$s in entire columns$\}$

$\mathcal{S}_X = \{\left[\begin{array}{c|c} A & B \end{array}\right] \in \mathcal{S}|$ adjacent rows of A are $1'$s$\}$

$\mathcal{S}_Z = \{\left[\begin{array}{c|c} A & B \end{array}\right] \in \mathcal{S}|$ adjacent columns of B are $1'$s$\}$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \in \mathcal{S}_X \qquad \left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right] \in \mathcal{S}_Z$$

Diagnose and Pseudo-

1 Initially in $+1$ eigenstate of elements of $\mathcal{S}_X$ and $\mathcal{S}_Z$

2 Error occurs

3 Measure elements of $\mathcal{S}_X$ and $\mathcal{S}_Z$

4 Eigenvalues of measurements allow diagnosis and fix modulo $\mathcal{T}$

X error:
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0 \end{array}\right]$$

Flips to -1 eigenvalues of
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&1&1&0\\ 0&0&0&0&0&1&1&0\\ 0&0&0&0&0&1&1&0\\ 0&0&0&0&0&1&1&0 \end{array}\right]$$

$$\left[\begin{array}{cccc|cccc} 0&0&0&0&1&1&0&0\\ 0&0&0&0&1&1&0&0\\ 0&0&0&0&1&1&0&0\\ 0&0&0&0&1&1&0&0 \end{array}\right]$$

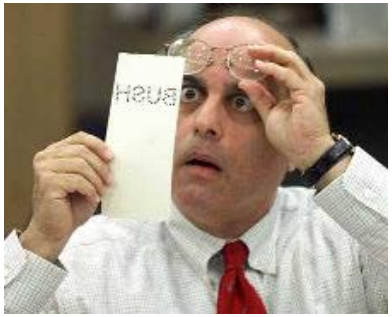
Apply fixing procedure
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0 \end{array}\right]$$

Cumulative error:
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0 \end{array}\right] \in \mathcal{T}$$

$$T = \bigoplus_{s \in S} T_s \otimes I$$

$$\rho_{err} \otimes |\psi\rangle\langle\psi|$$

Detailing The



$\mathcal{T} = \{c\left[\begin{array}{c|c} A & B \end{array}\right]$ | A has an even # of 1′s in columns

B has an even # of 1′s in rows$\}$

violate $\mathcal{T}$ condition

$$\begin{array}{cccc} e & e & o & e \end{array}$$

$$\left[\begin{array}{cccc|cccc} 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} o \\ e \leftarrow \text{violate } \mathcal{T} \text{ condition} \\ e \\ e \end{array}$$

syndrome:

$\mathcal{S}_X = \{\left[\begin{array}{c|c} A & B \end{array}\right] \in \mathcal{S}|$ adjacent rows of A are 1′s$\}$

fix by applying Z's consistent with fewest o's

fixes $\lfloor \frac{n}{2} \rfloor$-1 Z errors

Similar for $\mathcal{S}_Z$

Subsystem Quantum Error Correction

Codes are not really different from subspace codes, but the error recovery routine can be different.

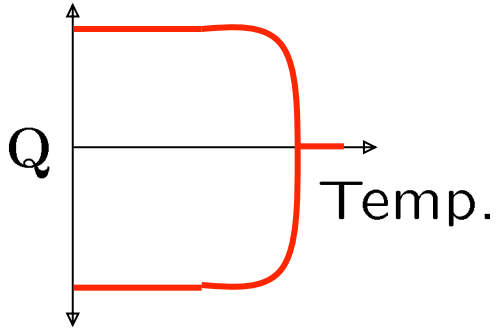
Does this lead to better codes?
(memory and fault-tolerant thresholds)

Properties of the Code

1. No memory threshold.

   Why? Number of correctible errors scales like linear dimension. Even “typical” errors (unlike toric code.)

2. Still, we can use subsystem code to define order parameter



this might be all right if the system is “self-correcting”

Quantum Compass Model

[B. Douçot, M. V. Feigel'man, L. B. Ioffe, and A. S. Ioselevich 05]

[Dourier, Becca, Mila 05]

$$H = -\sum_{i,j=1}^{n} J_X X_{i,j} X_{i+1,j} + J_Z Z_{i,j} Z_{i,j+1}$$

If this system has a gap, then we can encode quantum information in its degenerate ground state and it might be
protected at low enough temperature.

Is it gapped?

Green's Function Quantum Monte Carlo

powering method:

$$|\psi_L\rangle = (cI - H)^L |\psi_0\rangle$$

ground state:

$$|\psi_g\rangle = \lim_{L\to\infty} \frac{|\psi_L\rangle}{\sqrt{\langle\psi_L|\psi_L\rangle}}$$

Green's Function Quantum Monte Carlo

$$H = -\sum_{i,j=1}^{n} J_X X_{i,j} X_{i+1,j} + J_Z Z_{i,j} Z_{i,j+1}$$

powering method:

$$|\psi_L\rangle = (cI - H)^L |\psi_0\rangle$$

ground state:

$$|\psi_g\rangle = \lim_{L\to\infty} \frac{|\psi_L\rangle}{\sqrt{\langle\psi_L|\psi_L\rangle}}$$

assume $A = cI - H$ has all positive entries

decompose as

$$A = W M$$

diagonal positive

stochastic

Green's Function Quantum Monte Carlo

decompose as

$$A = W M$$

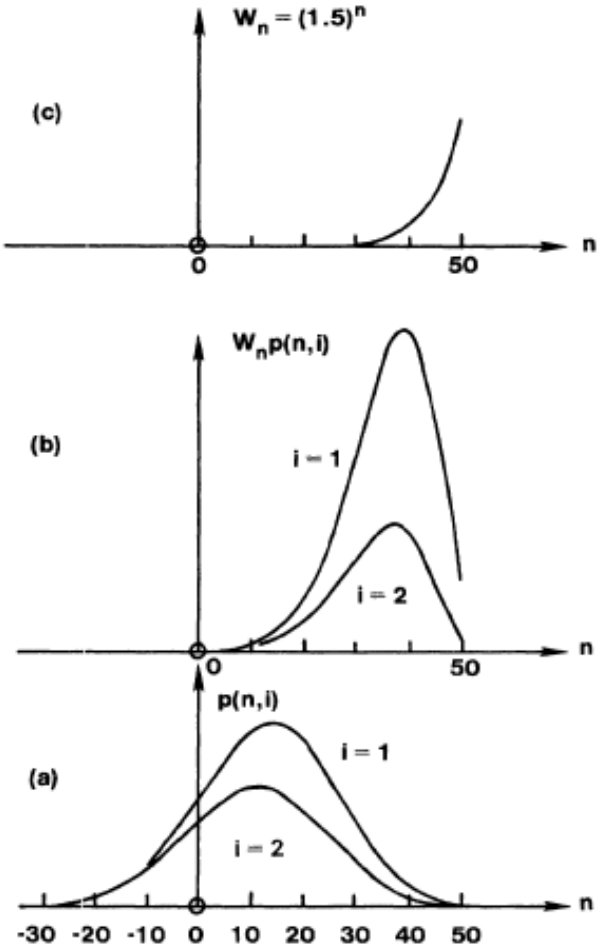
diagonal positive — stochastic

random walk, plus keeping track of weights

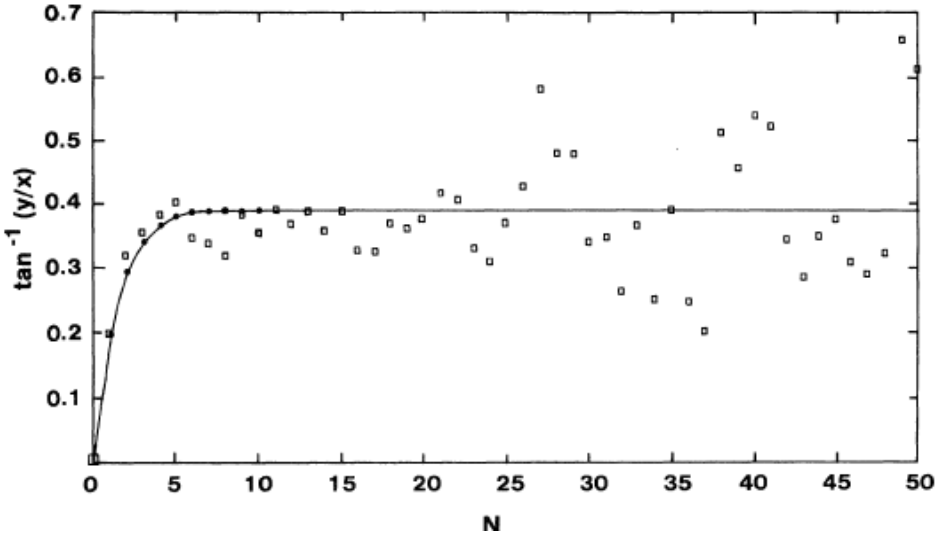
1. Choose new configuration according to *M*
2. Update weight depending on configuration

Green's Function Quantum Monte Carlo

Problem, while Markov process has nice convergence properties, the weights grow exponentially



[Hetherington 84]



[Hetherington 84]

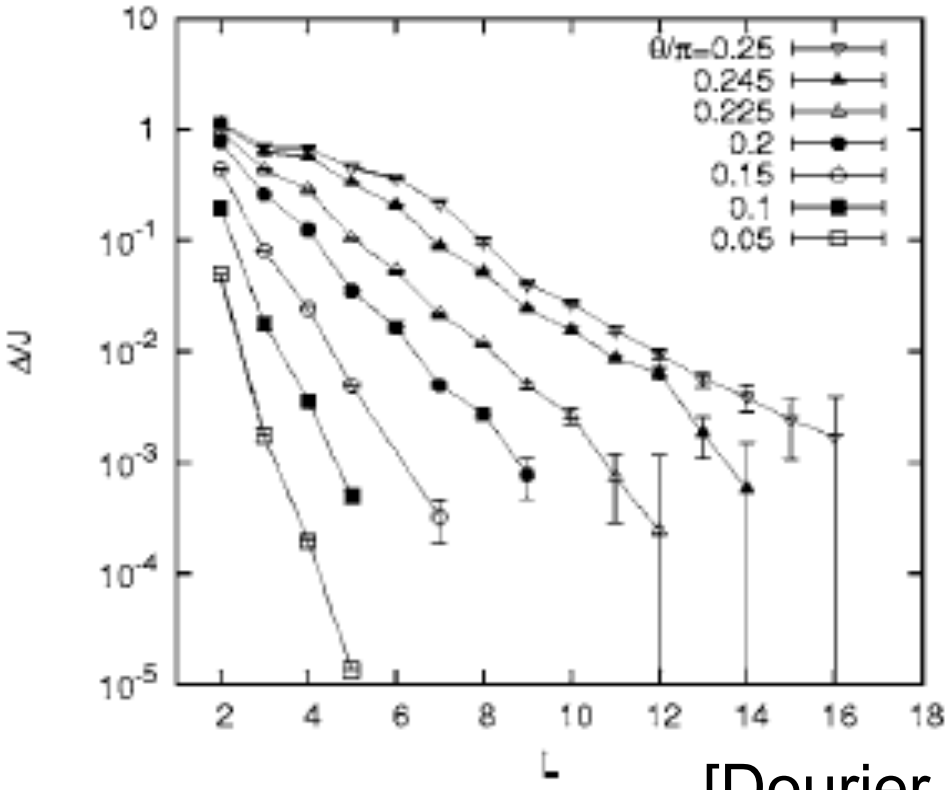
Green's Function Quantum Monte Carlo

One method for alleviating this problem is to work with multiple simultaneous walkers, and when the variance of the walkers grows too high, to couple the walkers and reset their weights to most important weights. [Buonaura and Sorella 98]

Also very useful is to use a guiding wavefunction

$$A'_{ij} = \frac{A_{ij}\phi_i}{\phi_j}$$

Quantum Compass Model [GFQMC]



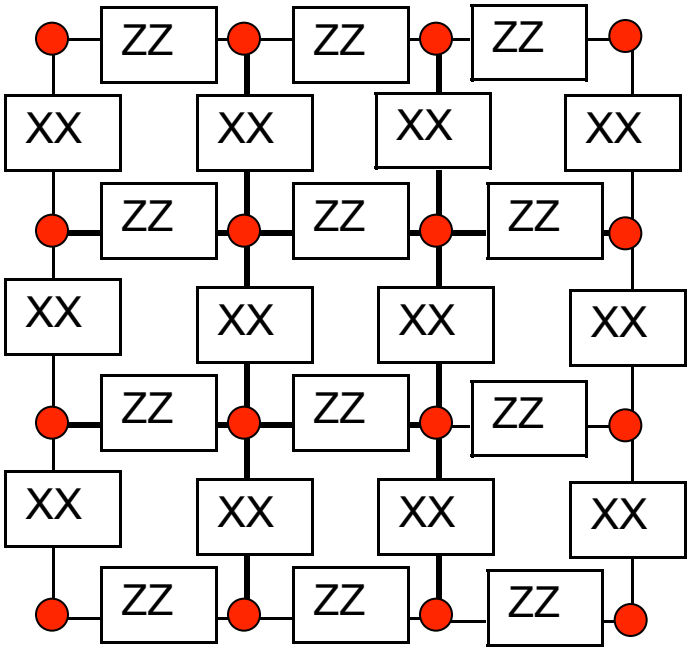
[Dourier, Becca, Mila 05]

$$H = -J \sum_{i,j=1}^{n} \cos(\theta) X_{i,j} X_{i+1,j} + \sin(\theta) Z_{i,j} Z_{i,j+1}$$

no gap in the thermodynamic limit

Quantum Compass Model

n by *n* square lattice

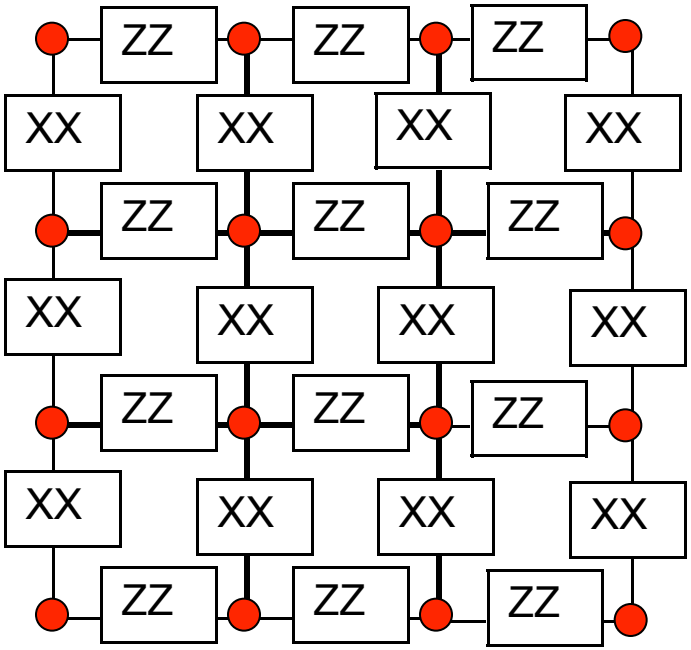


$$H = -\sum_{i,j=1}^{n} J_X X_{i,j} X_{i+1,j} + J_Z Z_{i,j} Z_{i,j+1}$$

Not very interested in this model in the first place, because excitations are probably point-like.

Beyond the QCM

n by *n* square lattice

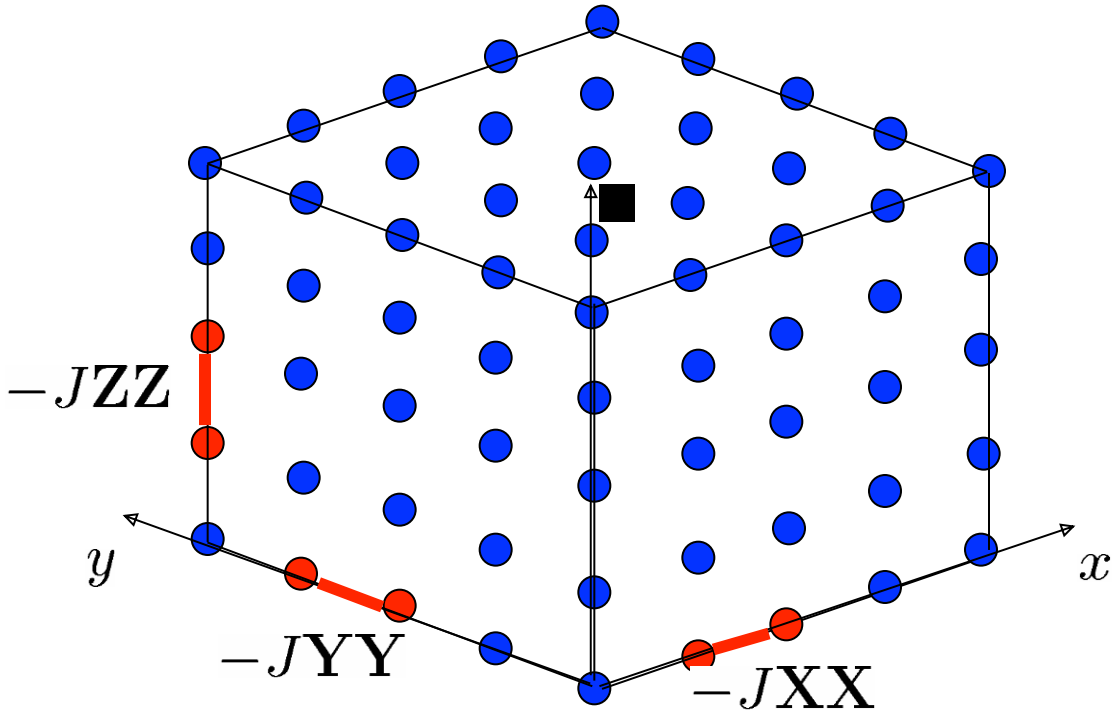


$$H = -\sum_{i,j=1}^{n}\sum_{k=1}^{n-1} J_X(k) X_{i,j} X_{i+k,j} + J_Z(k) Z_{i,j} Z_{i,j+k}$$

Long range interactions. May possibly protect ground state

Implementation in ion traps?

Back to the Third Dimension



$$H = -\sum_{ijk} J_X X_{i,j,k} X_{i+1,j,k} + J_Y Y_{i,j,k} Y_{i,j+1,k} + J_Z Z_{i,j,k} Z_{i,j,k+1}$$

Possible excitations string-like? Has “sign problem”!

Kitaev's Idea

Are there many-body quantum systems which can be used to robustly store quantum information and keep this information coherent during fault-tolerant computation with this information?

It would be disappointing that in order to verify Kitaev's idea we actually need to build the quantum computer.

Perhaps the new techniques described today can save the day?